

8	(a)	(i)	Let the power of the sun by P_{sun}. Considering the sun as a point source, the energy spreads out uniformly through space and the intensity is given by $\text{intensity} = \frac{P_{\text{sun}}}{4\pi r^2}$ $P_{\text{sun}} = 1400 \times \left[4\pi (150 \times 10^9)^2 \right]$ $= 4.0 \times 10^{26} \text{ W}$	M1 M1 A0
		(ii)	Unit of power = $\frac{\text{unit of work done}}{\text{unit of time}} = \frac{\text{kg m}^2 \text{ s}^{-2}}{\text{s}}$ $= \text{kg m}^2 \text{ s}^{-3}$ Unit of $(\sigma AT^4) = (\text{kg s}^{-3} \text{ K}^{-4})(\text{m}^2)(\text{K}^4)$ $= \text{kg m}^2 \text{ s}^{-3}$ Since the unit of P is the same as the unit of σAT^4, the equation is homogeneous.	B1 B1 A0
		(iii)	$P = \sigma AT^4$ $4.0 \times 10^{26} = (5.7 \times 10^{-8}) \left[4\pi (109 \times 6.37 \times 10^6)^2 \right] T^4$ $T = 5830 \text{ K}$	C1 A1
	(b)	(i)	Maximum speed is related to the maximum temperature Maximum temperature = $15 \times 10^6 \text{ K}$ Considering the protons to behave as ideal gases, KE of each proton = $\frac{3}{2} kT$ $\frac{1}{2} m_{\text{proton}} v^2 = \frac{3}{2} kT$ $\frac{1}{2} (1.67 \times 10^{-27}) v^2 = \frac{3}{2} (1.38 \times 10^{-23}) (15 \times 10^6)$ $v = 6.1 \times 10^5 \text{ m s}^{-1}$	C1 A1
		(ii)	As the proton approaches each other, their initial kinetic energies equal the electric potential energy at the closest distance. By conservation of energy, $\frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} = 2 \times \frac{1}{2} m_{\text{proton}} v^2$ $\frac{(1.6 \times 10^{-19})(1.6 \times 10^{-19})}{4\pi (8.85 \times 10^{-12}) r} = 2 \times \frac{1}{2} (1.67 \times 10^{-27}) (6.1 \times 10^5)^2$ $r = 3.7 \times 10^{-13} \text{ m}$	C1 A1

		(iii)	The radius of nucleus is of the <u>order of 10^{-15} m</u> . The <u>closest distance is much larger than the radius of the proton</u> and thus the two protons will be too far apart for fusion to take place.	B1 B1
		(iv)	When protons are very close to each other, the uncertainty in their position <u>Δx is small</u> . According to the Heisenberg Uncertainty Principle, this results in a <u>larger uncertainty in their momentum</u> . Thus, the large uncertainty in momentum Δp means that there is a significant probability that <u>some protons have much higher momentum (and thus kinetic energy) than the average</u> to overcome the Coulomb barrier in those instances, allowing fusion reaction to take place.	B1 B1
	(c)	(i)	Mass difference = $4(1.007276\text{u}) - 4.002603\text{u} = 0.026501\text{u}$ Energy release = $\Delta mc^2 = (0.026501)(1.66 \times 10^{-27})(3.0 \times 10^8)^2$ $= 3.96 \times 10^{-12}\text{J}$	C1 A1
		(ii)	Energy release by 0.70% of the mass $= (0.70/100)(4)(1.007276)(1.66 \times 10^{-27})(3.0 \times 10^8)^2$ $= 4.21 \times 10^{-12}\text{J}$ Since the two values can be rounded to $4 \times 10^{-12}\text{J}$, the two values are close to each other and the statement is valid. OR Percentage difference = $(4.21 \times 10^{-12} - 3.96 \times 10^{-12}) / (3.96 \times 10^{-12}) = 6.3\%$ Since the percentage difference between the two values is less than 10%, the statement is valid.	M1 A1 (M1) (A1)
	(d)	(i)	$\gamma = \frac{\sqrt{2m(U-E)}}{\hbar} = \frac{\sqrt{2(1.67 \times 10^{-27})(10^{-13} - 10^{-16})}}{6.63 \times 10^{-34} / 2\pi} = 1.731 \times 10^{14}$ $P \approx e^{-2\gamma d} = e^{-2(1.731 \times 10^{14})(1.8 \times 10^{-13})} = 8.6 \times 10^{-28}$	C1 A1
		(ii)	The <u>vast number of protons</u> in the Sun of the order of more than 10^{38} <u>suggests some protons will be able to tunnel through the potential barrier</u> for fusion to occur.	B1